

[Home](#) > [Research Perspectives on Software Engineering and Systems Design](#) > Conference paper

Hop Count-Based Detection Scheme for Sinkhole Attack

| Conference paper | First Online: 01 September 2025

| pp 1–16 | [Cite this conference paper](#)

 [Save conference paper](#)

[View saved research](#) >



[Research Perspectives on Software Engineering and Systems Design](#)
(CoMeSySo 2024)

[L. C. Sejaphala](#) , [V. Malele](#) & [F. Lugayizi](#)

 Part of the book series: [Lecture Notes in Networks and Systems](#) ((LNNS, volume 1491))

 Included in the following conference series:
[Proceedings of the Computational Methods in Systems and Software](#)

 230 Accesses  1 Citation

Abstract

Next Generation Wireless Sensor Network is a technology that allows various devices to collect data through sensors or sensor networks and share it in an Internet environment. Because most communication is wireless, the network is highly susceptible to various attacks such as blackholes, selective forwarding, and sinkhole attacks, which can compromise the integrity and reliability of the data. Among these, sinkhole attacks are particularly harmful to networks. However, detecting sinkhole attacks in 5G-WSN is more challenging than in ad-hoc networks and wireless sensor networks owing to environmental differences. 5G-WSN have found widespread applications in various fields such as road traffic monitoring, military surveillance, eHealth, mining, manufacturing, and more. These networks play a crucial role in collecting and transmitting critical data, and any compromise in their integrity can have severe consequences, particularly in applications that impact human lives. Therefore, ensuring the security and reliability of these networks is of paramount importance to safeguard the lives and interests of people involved in these applications. This study proposed a Hop Count-Based Detection Scheme for Sinkhole Attack to mitigate the destructive sinkhole attack in the 5G-WSNs. The scheme uses hop counts of the nodes as the bases of detecting the sinkhole nodes in the network. The performance of the proposed scheme was evaluated in terms of probability of false positive, probability of false negative, and probability of detection in contrast to Ibrahim's Hop Count-Based Detection Algorithm. Moreover, the study evaluated the effects of the packet dropping sinkhole attack with regard to packets loss ratio, packet delivery ratio. On average, from the simulation results, the proposed scheme outperformed Ibrahim's detection algorithm.

 This is a preview of subscription content, [log in via an institution](#)  to check access.

Access this chapter

Log in via an institution →

Subscribe and save

 **Springer+**

from €37.37 /Month

- Starting from 10 chapters or articles per month
- Access and download chapters and articles from more than 300k books and 2,500 journals
- Cancel anytime

[View plans](#) →

Buy Now

^ Chapter

EUR 29.95

Price includes VAT (Taiwan (R.O.C.))

- Available as PDF
- Read on any device
- Instant download
- Own it forever

[Buy Chapter](#) →

▼ eBook

EUR 181.89

▼ Softcover Book

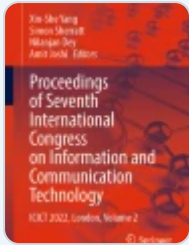
EUR 219.99

Tax calculation will be finalised at checkout

Purchases are for personal use only

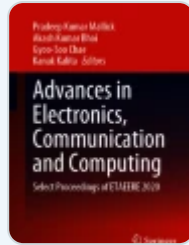
[Institutional subscriptions](#) →

Similar content being viewed by others



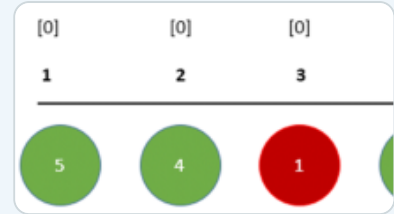
Investigating the Sinkhole Attack in Cognitive Wireless Sensor Network

Chapter | © 2023



Using Ad Hoc on Demand Distance Vector Technique for Identifying Sinkhole...

Chapter | © 2021



The Design of a Defense Mechanism to Mitigate Sinkhole Attack in Software Defined...

Article | Open access
13 April 2020

Explore related subjects

Discover the latest articles, books and news in related subjects, suggested using machine learning.

[Algorithms](#)

[Internet of Things](#)

[Motion Detection](#)

[Mobile and Network Security](#)

[Network topology](#)

[Wireless and Mobile Communication](#)

[Energy-Efficient Protocols in Wireless Sensor Networks](#)

References

1. Chourabi, H., Nam, T., Walker, S., Gil-Garcia, J.R., Mellouli, S., Nahon, K., Pardo, T.A., Scholl, H.J.: Understanding smart cities: an integrative framework. In: Book Understanding Smart Cities: An Integrative Framework, pp. 2289–2297 (2012)

[Google Scholar](#)

2. Al-Fuqaha, A., Guizani, M., Mohammadi, M., Aledhari, M., Ayyash, M.: Internet of things: a survey on enabling technologies, protocols, and applications. *IEEE Commun Surv Tutor.* **17**(4), 2347–2376 (2015)

[Article](#) [Google Scholar](#)

3. Eswaran, S., Honnavalli, P.: Private 5G networks: a survey on enabling technologies, deployment models, use cases and research directions. *Telecommun. Syst.* **82**(1), 3–26 (2023)

[Article](#) [Google Scholar](#)

4. Thompson, J., Ge, X., Wu, H.C., Irmer, R., Jiang, H., Fettweis, G., Alamouti, S.: 5G wireless communication systems: prospects and challenges [Guest Editorial]. *IEEE Commun Mag.* **52**(2), 62–64 (2014)

[Article](#) [Google Scholar](#)

5. Omar, A.A., Soudan, B., Altaweel, A.: A comprehensive survey on detection of sinkhole attack in routing over low power and lossy network for internet of things. In: *Internet of Things* (2023)

[Google Scholar](#)

6. Abdullah, I., Rahman, M.M., Roy, M.C.: Detecting sinkhole attacks in wireless sensor network using hop count. *Int. J. Comput. Netw. Inf. Secur.* **7**, 50–56 (2015)

[Google Scholar](#)

7. Ali, M., Nadeem, M., Siddique, A., Ahmad, S., Ijaz, A.: Addressing sinkhole attacks in wireless sensor networks—a review. *Int. J. Sci. Technol. Res.* **9**, 406–411 (2020)

[Google Scholar](#)

8. Vamsi, P.R., Kant, K.: Secure data aggregation and intrusion detection in wireless sensor networks. In: Book Secure Data Aggregation and Intrusion Detection in Wireless Sensor Networks, pp. 127–131 (2015)

[Google Scholar](#)

9. An, G., Cho, T.H.: Improving sinkhole attack detection rate through knowledge-based specification rule for a sinkhole attack intrusion detection technique of IoT. *Int. J. Comput. Netw. Appl.* **9**, 169–178 (2022)

[Google Scholar](#)

10. Babaeer, H.A., Al-Ahmadi, S.A.: Efficient and secure data transmission and sinkhole detection in a multi-clustering wireless sensor network based on homomorphic encryption and watermarking. *IEEE Access.* **8**, 92098–92109 (2020)

[Google Scholar](#)

11. Kori, S.M., Krishnamurthy, G.N., Sidnal, N.S.: Distributed wormhole attack mitigation technique in WSNs. *Int. J. Comput. Netw. Inf. Secur.* **11**(5), 20–27 (2019)

[Google Scholar](#)

12. Shahid, H., Ashraf, H., Ullah, A., Band, S.S., Elnaffar, S.: Wormhole attack mitigation strategies and their impact on wireless sensor network performance: a literature survey. *Int. J. Commun. Syst.* **35** (2022)

[Google Scholar](#)

13. Salehi, S.A., Razzaque, M.A., Naraei, P., Farrokhtala, A.: Detection of sinkhole attack in wireless sensor networks. In: Book Detection of Sinkhole Attack in Wireless Sensor Networks, pp. 361–365 (2013)

[Google Scholar](#)

14. Waoo, D.A.A., Tiwari, V.K.: Challenges in sinkhole attack detection in wireless sensor network. *Indian J. Data Commun. Netw.* **1**(4), 1–7 (2021)

[Article](#) [Google Scholar](#)

15. Hu, H., Chang, W.: On the mitigation of controllable event triggering attack in WSNs. In: 2020 29th International Conference on Computer Communications and Networks (ICCCN), pp. 1–6 (2020)

[Google Scholar](#)

16. Kibirige, G., Sanga, C.: A survey on detection of sinkhole attack in wireless sensor network. *Int. J. Comput. Sci. Inf. Secur.* **13**, 1–9 (2015)

[Google Scholar](#)

17. Subba, B., Biswas, S., Karmakar, S.: A game theory based multi layered intrusion detection framework for wireless sensor networks. *Int. J. Wireless Inf. Netw.* **25**, 399–421 (2018)

[Article](#) [Google Scholar](#)

18. Jamshidi, M., Poor, S.S.A., Qader, N.N., Esnaashari, M., Mohammad, R.M.: A lightweight algorithm against replica node attack in mobile wireless sensor networks using learning agents. *IEIE Trans. Smart Process. Comput.* **8**(1), 58–70 (2019)

[Article](#) [Google Scholar](#)

19. Gayathri, V., Kumarswamy, Y.S.: Routing and neighbor node cooperation scheme to achieve energy efficient encryption for integrating wireless sensor networks with internet. In: Book Routing and Neighbor Node Cooperation Scheme to Achieve Energy Efficient Encryption for Integrating Wireless Sensor Networks with Internet (2019)

20. Jebadurai, I.J.R., Rajsingh, E.B., Paulraj, G.J.L.: A novel node collusion method for isolating sinkhole nodes in mobile Ad Hoc cloud. In: Book A Novel Node Collusion Method for Isolating Sinkhole Nodes in Mobile Ad Hoc Cloud (2018)

[Google Scholar](#)

21. Bahekmat, M., Yaghmaee, M.H., Sadat, A., Yazdi, H., Sadeghi, S.: A novel algorithm for detecting sinkhole attacks in WSNs. Int. J. Comput. Theory Eng., 418–421 (2012)

[Google Scholar](#)

Author information

Authors and Affiliations

North-West University, Vanderbijlpark, Gauteng, South Africa

L. C. Sejaphala & V. Malele

North-West University, Mafikeng, North West, South Africa

F. Lugayizi

Corresponding author

Correspondence to [L. C. Sejaphala](#) .

Editor information

Editors and Affiliations

Faculty of Applied Informatics, Tomas Bata University in Zlín, Zlín, Czech Republic

Radek Silhavy

Faculty of Applied Informatics, Tomas Bata University in Zlín, Zlín, Czech Republic

Petr Silhavy

Rights and permissions

[Reprints and permissions](#)

Copyright information

© 2025 The Author(s), under exclusive license to Springer Nature Switzerland AG

About this paper

Cite this paper

Sejaphala, L.C., Malele, V., Lugayizi, F. (2025). Hop Count–Based Detection Scheme for Sinkhole Attack. In: Silhavy, R., Silhavy, P. (eds) Research Perspectives on Software Engineering and Systems Design. CoMeSySo 2024. Lecture Notes in Networks and Systems, vol 1491. Springer, Cham. https://doi.org/10.1007/978-3-031-96380-3_1

[.RIS](#) [.ENW](#) [.BIB](#)

DOI	Published	Publisher Name
https://doi.org/10.1007/978-3-031-96380-3_1	01 September 2025	Springer, Cham
Print ISBN	Online ISBN	eBook Packages
978-3-031-96379-7	978-3-031-96380-3	Intelligent Technologies and Robotics
		Intelligent Technologies and Robotics (R0)
		Springer Nature
		Proceedings excluding
		Computer Science

Keywords

- 5G
- WSN
- Sinkhole attack
- IoT
- and Hop-Count

Publish with us

[Policies and ethics](#) 

